

Parallel Computing Lab 1: The Amdahl Reality Gap

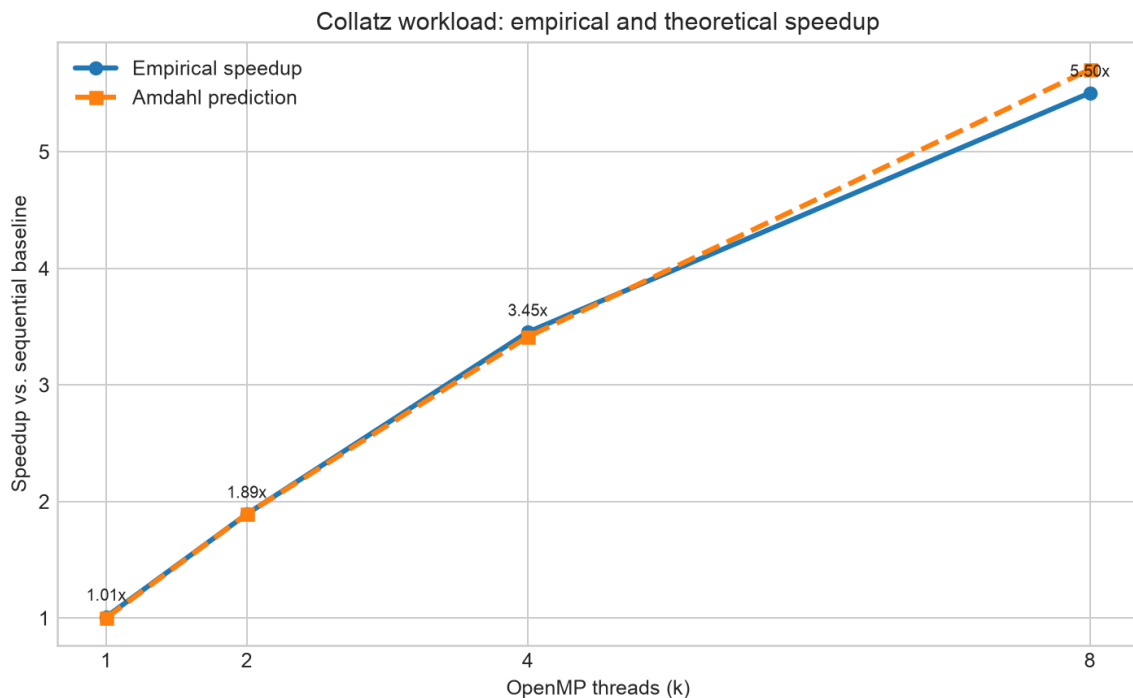
Empirical results from Apple M3, macOS (N = 13,379,000)

Method

The program was compiled with clang 16, OpenMP libomp 23.1.2, and -O2. Run 1 was treated as warm-up; every displayed mean averages runs 2 and 3. All runs produced max stopping time 688, checksum 117734093, and 10,796,034 values above 100 steps.

Scaling results

k	Avg runtime (s)	S_emp	S_theo	Reality gap
1	1.631344	1.0100	1.0000	-0.0100
2	0.871094	1.8914	1.8914	0.0000
4	0.477274	3.4521	3.4123	-0.0398
8	0.299301	5.5048	5.7067	0.2019



Q1. Micro-architectural root cause of false sharing

The naive variant writes `hits[tid]` from each worker. Although the array elements are logically separate, adjacent `uint64_t` counters occupy the same 64-byte cache line. Each write therefore requires exclusive ownership of that line, repeatedly invalidating peer copies and creating coherence-fabric traffic. This is the practical

MESI/MOESI-style invalidation problem the exercise targets (Apple does not publicly document its exact protocol). At 8 threads the naive mean was 0.345444 s, versus 0.304796 s with OpenMP reduction: a 1.1334x time penalty. Reduction keeps accumulation private until combination and avoids repeated contended writes.

Q2. Physical cores and SMT

The machine reports 8 logical CPUs and 8 physical CPUs: Apple M3 has four performance and four efficiency cores, not SMT/Hyper-Threading pairs. Therefore $k=8$ is not a move from physical cores to SMT siblings. Speedup still is not linear: it rises from 3.4521x at $k=4$ to 5.5048x at $k=8$, because the added efficiency cores have lower per-core throughput and the workload also pays scheduling, synchronization, cache, and memory-system costs. On an SMT machine, front-end, execution units, caches, TLBs, and memory bandwidth are shared between sibling hardware threads; that is not the limiting topology measured here.

Q3. Empirical versus theoretical Amdahl discrepancy

From the $k=2$ result, $p = 2[1 - 1/S_{\text{emp}}(2)] = 0.942593$. The fitted Amdahl curve predicts 5.7067x at $k=8$, while the measurement is 5.5048x, leaving a 0.2019x reality gap. Amdahl's ideal form omits (1) cache-coherence and memory-bandwidth contention, and (2) OpenMP runtime costs such as thread-team coordination, work distribution, and the final reduction/barrier. This Apple M3 also mixes performance and efficiency cores, so its workers are not uniform execution units.

Q4. Scheduling trade-off

At 8 threads, static averaged 0.302276 s; static,1000 0.304546 s; dynamic,100 0.284260 s; dynamic,10000 0.287156 s; and guided 0.281572 s. The imbalance in Collatz trajectories made adaptive scheduling worthwhile on this machine. Among the tested chunks, lock-queue contention did not outweigh the balancing benefit: dynamic,100 was faster than static and dynamic,10000. The data do not justify claiming a crossover chunk size not tested; guided was the fastest tested policy.

Reproducibility

The source code, raw timing log, machine specification, and PNG plot are packaged alongside this report. Rebuild with: `clang -O2 -Xpreprocessor -fopenmp -I/opt/homebrew/opt/libomp/include -L/opt/homebrew/opt/libomp/lib -Wl,-rpath,/opt/homebrew/opt/libomp/lib -lomp -std=c11 collatz.c -o collatz`